

CTEV

A 15 months old male child Mr. Raj from TVR, brought by mother, whose reliability is good, came c/o, - deformity in (R) leg since birth.

- H/OPI: The child was identified to have deformity at birth.
- After 15 days, POP (below knee cast) was applied to both legs for 3 months.
 - After 3 months, (L) side was corrected. Only little improvement on (R) leg. Then he was advised CTEV splint.
 - Now, the child is referred here.

Past H/o:

- Non-consanguineous marriage.

AN H/o: - No H/o oligohydramnios.

- No H/o TORCH infections

- No H/o multiple pregnancy.

Natal H/o: FTNVD, 1st child.

Developmental H/o: Attained as per age.

Immunisation H/o: Immunised upto age.

Family H/o: No significant family history.

General Examination:

- Alert and awake

No PICCLE.

No kyphosis/scoliosis.

Spine (N).

Hip joint (N).

Vitals:

Inspection:

Attitude: Hip: Neutral.

Knee: flexed

Ankle: Equinus, plantar flexion.

- Subtalar joint : Varus - inverted.
- Talo-navicular joint : Adducted.
- Medial border : Concave, deep crease present.
- Lateral border : Convex, stretched, callusities on lateral malleolus.
- Lateral malleolus } prominent.
Head of talus }

Palpation:

- Not warm, no tenderness.
- Inspectory findings are confirmed.
- Deformity (flexible / rigid).

Movements:

Ankle : PF : 40° .

DF : restricted / not possible.

Subtalar : Inversion - 30° .

Eversion - restricted.

Talonavicular :

Abduction -

Adduction - positive.

Left leg : (N), no deformity

Adequate correction $\bar{c}$ 10° DF.

Measurements:

Thigh

(R)

(L).

17 cms

17 cms.

Leg

15 cms

17 cms.

- 2 cms shortening on (R) leg segment ———.
- Distal neurovascular status (N).
- No LNopathy.
- Other joints (N), other limb (N).

Diagnosis:

A case of partially corrected CTEV on (R) foot with shortening of leg.

CHRONIC OSTEOMYELITIS

A 77 years old Mr. Kalyan from TVR, came c/o
- discharge from upper part of front of leg x 2 yrs.

HPI:

- He was apparently (N) before 2 yrs, then he sustained a trauma of (R) leg for which he was operated.
- After 1 month, he noticed throbbing pain & discharge at the site of operation.
- Discharge (was oozing out from Sx site).
 - continuous, purulent, scanty discharge for 2 yrs.
 - not foul smelling, not blood stained.
 - H/o bony spicules in discharge.
 - pain aggravated on walking, relieved by rest.

Past H/o:

- Known HT.
- N/K/c/o DM, TB, epilepsy, cardiac disorders.

Personal H/o:

- Mixed diet.
- (Bowel & bladder habits (N) x).
- Alcohol, smoking.

Rx H/o:

General Examination:

- conscious, oriented, moderately built & nourished.
- No PICCLE.
- pedal edema diffuse from foot to knee.
- skin hyperpigmented & thickened over front of legs.
- smooth surface.
- no apparent muscle wasting / limb shortening.

Vitals: PR, RR, BP.

Examination of R leg:

After getting consent from pt, he is in supine position.

Attitude: Hip: neutral.
Knee: Extended.
Ankle: neutral.

Inspection:

- 4 sinuses present on (A) surface of upper 1/3rd of fibula (tibia).
- 1 → actively discharging ; 3 → healed lesions.
- Actively discharging sinus of size 0.5 cms, healthy granulation tissue at the base of ulcer.
- Discharge is scanty, not foul smelling, not blood stained, seropurulent.

Palpation:

- Not warm, no tenderness.
- All inspectory findings are confirmed.
- ~~sinus~~ ^{sinus} ~~not~~ adherent to underlying bone.
- granulation tissue ⊕, does not bleed on touch.
- on moving the # site, no abnormal mobility.

→ If bone palpable ⇒ bone broadened, thickened (X-ray - smooth callus)
X-ray - granular spot due to irreg. periosteal reaction

Measurements:

- No limb length discrepancy.

Movement:

- Knee joint $\left\{ \begin{array}{l} \text{Flexion: } 0^\circ - 140^\circ \\ \text{Extension: } 140^\circ - 0^\circ \end{array} \right\}$ Normal range of mvs.

- Ankle joint → (N).
- Distal neurovascular status (N).
- No LN enlargement.
- No adjacent joint stiffness.
- Opposite limb: (N).

Diagnosis:

A case of post ^{traumatic} (operative) chronic osteomyelitis involving proximal 1/3rd of tibia.

DD: Implant infection
TB OM.
Soft tissue infxn.
Ewing's sarcoma.

Inv: ① X-ray

- Thickening
- Sclerosis (patchy).
- Sequestrum (dense area, deep²ly).
- Involucrum, cloaca.

② Sinogram.

③ CT, MRI.

④ Blood (ESR↑).

⑤ Pus culture.

R: ① Sequestrectomy

② Saucerisation.

SS's ③ Saline irrigation

④ Skin graft

⑤ Skeletal stabilisation

Complications:

(★) Acute exacerbation

(★) Growth abnormalities.
(Shortening, lengthening,
deformity)

(★) Pathological #

(★) Sinus tract malignancy. (SCC).

(★) Amyloidosis (rare).

NON-UNION

A 38 years old Mr. Raja from TVR came c/o

- deformity of R UL for 1 month.
- pain during movement of limb for 1 month.

H/OI:

→ He was apparently N before 12 years. Then he sustained
R UL, for which he was R in TMC.
→ Pt. was apparently N before 1 month, then he noticed a
deformity R UL, which was associated c/o

- pain

pricking type

radiating towards R hand

aggravated by mvts, & relieved by rest.

- H/o difficulty in using R arm for past 1 month.
- H/o abnormal mobility.
- No H/o fever
- No H/o open wound
- No H/o discharge
- No H/o ⁱⁿtracutaneous R / oil massage.

Past H/o:

- N/K/c/o DM, HT, TB, epilepsy, (cardiac disorders)
- No H/o prev. Sx

Personal H/o:

- Mixed diet
- Not alcoholic & smoker.

General Examination: Cons., oriented, mod B & N, no PICCLE, no deformities

Examination of UL: Vitals: PR, RR, BP.

- After getting consent, on exposure from shoulder, sitting

Position

● Attitude: Shoulder - slightly abducted.
Elbow - flexed, mid prone position.
Wrist - neutral.

Carrying angle: In extended & supination of elbow - ---
(N): $11^{\circ} - 15^{\circ}$

Inspection:

- A diffuse swelling & deformity in (R) side, extends from middle 1/3rd of arm, just above the elbow.
- A vertical scar is present.
- No ulcer / sinus over the skin.
- No wasting of muscles.
- Apparent shortening of (R) limb.

Palpation:

- Not warm, not tender.
- A deformity in lower end of (R) arm, size $16 \times 9 \text{ cms}$, irregular in shape, firm consistency, abnormal mobility present, no crepitus.
- Three point relationship of joints (N).
- Stability of elbow present.

Movements:

Active: At shoulder joint → (N).

Elbow joint → Restriction of flexion & extension (+).

Supination & pronation: (N).

Passive: Flex^{can} upto 90° at elbow jt.

Measurements:

	(R)	(L)
→ Arm	25 cms	29 cms
Forearm	15 cms	15 cms.

- Shortening of 4 cms in (R) arm present.
- Distal neurovascular status (N).
- No LN. Enlargement.
- Other jts & limbs: (N).

Diagnosis:

A case of post-traumatic non-union of R distal

humerus & shortening of R arm.

Inv: X-ray (obliterated medullary cavity, smooth & sclerotic # lines).

- Rx:
- ① Open reduction, internal fixation & bone grafting.
 - ② Excision of fragments (& replacement by artificial moulds).
 - ③ Ilizarov's external fixator (rare).
 - ④ If asymptomatic, no Rx.

PERIPHERAL NERVE INJURIES

A 46 years old Mrs. Latha coming from Thiruvananthapuram. of low SES, homemaker, came with complaints of

Radial N: - Inability to hold (or) carry objects for _____ days.
(wrist jt).

Ulnar & Median N: - Inability to grasp objects for _____
(fingers)

Common peroneal N: - Inability to lift the foot for _____

History of Presenting Illness:

Patient was apparently normal before _____, following which she sustained a trauma, then she developed,

→ inability to hold, carry objects / -inability to use finger /

Inability to lift foot (or) climb stairs.

H/o numbness & paresthesia present over _____ area.

H/o skin changes (dry, wrinkled, moist)

H/o hypopigmented patches (leprosy)

Past H/o:

No H/o similar illness in the past.

No H/o DM, HT, leprosy, syphilis.

Personal H/o:

No H/o smoking

No H/o alcohol intake.

Family H/o:

No H/o any neurological illness in the family.

General Examination:

Patient is conscious, oriented.

Moderately built & nourished.

No PICC LE.

Neurocutaneous markers

Attitude:

Shoulder joint: Slightly abducted.

Elbow: Slightly flexed, midprone position.

Wrist: Neutral (wrist drop / flexed / extended).

for foot, Hip: neutral

Knee: Extended.

Foot: plantar flexion (foot drop).
Ankle

Deformity:

→ Radial N: Unable to extend the wrist.

Unable to extend MCP jt.

& flex IP jt.

Ulnar deviation of hand present.

Thumb at adduction.

Ulnar N: → Unable to flex MCP jt

- Unable to extend IP jt

- ~~no~~ hollowers seen between metacarpals.

} of little & ring finger.

Median N: → Unable to flex MCP jt

- Unable to extend IP jt

} of index and middle finger.

Common peroneal N: Unable to do dorsiflexion of foot.

→ No skin changes.

Palpation:

Not warm

Not tender

Inspection findings are confirmed.

Radial Nerve:

Motor Exam:

Triceps → power not elicited

Brachioradialis → power 3/5.

Extensor of wrist → Grade 2 power (2/5).

Extensor digitorum → unable to extend MCP joint.

Extensor pollicis longus → Unable to extend distal IP jt, while catching hold of the proximal IP joint of thumb.

Sensory Exam: (pain, touch, temp, pressure).

Sensory loss over anatomical snuff box.

Ulnar Nerve:

Motor Exam:

Flexor carpi ulnaris: Tendon is felt over pisiform bone on giving resistance over ulnar borders of hand.

Abductor digiti minimi: Able to abduct little finger against resistance (5/5).

Palmar interossei: Card test: able to hold the card.

Dorsal interossei: able to abduct finger against resistance.

Adductor pollicis: Froment's sign negative.

Lumbricals (Medial 2): Flexion at MCP jts.

Extension at IP jts.

FDP of medial 2 fingers: Flexion at DIP jts possible.

Sensory Exam:

Sensation intact over little finger (tip).

Median Nerve:

Motor Exam:

Flexor carpi radialis: Able to flex the wrist against resistance over radial border of hand.

Opponens pollicis - able to oppose fingers (thumb).

FPL: able to flex the terminal phalanx of thumb against resistance.

FDS & FDP: oschner's clasping test (lat. one third of FDP).

Abductor pollicis brevis: pen test.

Sensory Exam:

Sensation over index finger volar aspect preserved.

Reflex:

Triceps

Brachioradialis.

Common peroneal Nerve:

Motor Exam:

Sensory Exam:

Measurements:

Arm

Forearm

→ Distal neurovascular status: normal (radial pulse).

Diagnosis :

A case of post-traumatic radial/ulnar/median/
common peroneal nerve palsy of left/right limb with
wrist drop / partial claw hand / foot drop.

Investigations :

(*) Blood

(*) Nerve conduction study

(*) EMG.

Treatment :

Deformity - Cock up splint / Knuckle bender splint.

Muscle atrophy - Electrical stimuli.

Joint stiffness - Joint mobilisation / physioRx.

Tendon Transfer :

Pronator teres to ECRB → wrist drop.

FCR to ED → Finger drop.

APL to EPL → Thumb drop.

OSTEO ARTHRITIS

Mr. Ramadas, 65 years old, coming from Mayiladuthurai, male farmer by occupation, came with chief complaints of,

- pain and stiffness over R knee joint for past 3 months.
- swelling over R knee jt for past 4 months.

History of Presenting Illness:

He was apparently normal before 4 months, after which he developed

- swelling which was gradual in onset, present continuously.

H/o pain for past 3 months, which was

gradual in onset
progressive in nature
unilateral
dull aching pain
continuous
aggravated by movements
relieved by rest.

H/o stiffness present for past 3 months.

- aggravated on sitting for a long duration
- no early morning stiffness present.
- relieved after walking.

H/o difficulty in squatting

H/o difficulty in sitting & crossed legs.

H/o difficulty in climbing stairs.

H/o inability to bear weight on affected side.

H/o restricted movements.

H/o crepitus on movement.

H/o trauma.

H/o fever.

H/o night pain.

H/o pain & swelling in any other joint.

Past History:

No H/o previous # / trauma in that joint.

No H/o DM, HT, TB, epilepsy.

No H/o surgery.

Personal History:

No H/o smoking, alcohol intake.

Family History:

No relevant family history (autoimmune disease, TB)

General Examination:

Conscious, oriented, Mod. built & nourished.

No PICCLE.

No skin lesion (psoriatic arthritis)

Local Examination of Knee Joint:

After getting consent, patient in supine position.

Inspection:

Attitude:

Hip:

neutral

Knee:

Extended / fixed
flexion
deformity
present.

Ankle:

neutral

Ⓐ

neutral.

Extended

neutral.

Deformity:

- Fixed flexion deformity / genu varum.

- Fullness present in the suprapatellar fossa.
- No scar, sinuses, ulcers, dilated veins.
- No obvious muscle wasting present.
- No shortening on inspection.
- Patient in prone position: popliteal fossa normal.

Palpation:

(R)

(L)

Not warm

- Not warm
- Tenderness present over medial & lateral joint line.
- (On moving patellofemoral jt, no tenderness, crepitus present.)
- Fluctuation positive
- Patellar tap present
- Synovial thickening present/not
- Crepitus present on flexion/extension of knee jt.
- No other swelling around the joint.

[Patient in prone position,

- Monart Baker cyst.
- On flexion less prominent, extension - prominent.
- Compressible
- Transillumination +ve.]
- = Centrally placed.

If Baker cyst present.

Movements:

Knee joint: Flexion $\Rightarrow 0 - 140^\circ$.

Flexed knee jt: Internal rotation: upto 30° .

External rotation: upto 15° .

- No abnormal mobility
- No locking
- Fixed flexion deformity +/-.

Measurements:

Muscle mass: Thigh - 28 cms

Leg - 20 cms

Limb length: Thigh - 32 cms

Leg - 28 cms

} Limb length discrepancy +/-.

- No distal neurovascular deficit
- No lymph node enlargement
- Antalgic gait present
- Other limb normal
- Adjacent joints normal

Diagnosis:

B/L / U/L osteoarthritis of right / left limb
with / without, FFD / genu varum deformity.

Investigations:

X-ray: Narrowing of joint space.

Subchondral cysts.

Osteophytes.

Loose bodies.

Blood tests (ESR, TC, DC).

Serum uric acid (To R/O gout).

Arthroscopy

Treatment :

Medical Rx :

- (★) Analgesics (Diclofenac, Cox-2 Inhibitors, Morphine).
- (★) Chondroprotective agents (Chondroitin $\Rightarrow$ rarely used, Sulphate), Glucosamine.
- (★) Viscosupplements (rare).
- (★) Weight reduction
- (★) Avoid stress & strain to joint.
- (★) Local heat - pain relief.
- (★) Exercises - Quadriceps strengthening.

Surgical Rx :

- (★) High tibial osteotomy (mild cases & for genu varum)
- (★) Total knee replacement (severe cases).

ORTHO CASE SHEETS

MALUNION

A 65 years old female Mrs. Kalyani from Seerkali, house wife by occupation came c/o.

- bending of (R) leg for 20 days.

H/OI:

She was apparently normal for 2 months, then while she was walking, accidentally fell down,

- developed pain & swelling in the middle of (R) leg,

lasted for 1 week, treated c/o analgesics, (subsided in 4 days).

- Before 20 days, while walking, her leg suddenly bent outward. c/o pain, not radiating, dull aching, aggravated on walking.

- H/o difficulty to stand & walk on her own, but able to stand c/o support.

- H/o restriction of movements at ankle joint.

- No H/o native Rx.

- No H/o oil massage

- No H/o operative interventions.

- No H/o fever, discharge, open wounds.

Past H/o:

- No H/o similar illness in the past.

- N/K/C/O DM, TB, BA, epilepsy

- No H/o drug intake.

- No H/o previous Sx.

Personal H/o:

- She takes mixed diet.

- (Bowel & bladder habits (N)) x

- Alcohol, smoking.

R H/o:

- She was admitted here before 9 days L.R.

R: Below knee cast was applied.

General Examination:

- Conscious, oriented, moderately built & nourished.

- No PICCLE.

Vitals: BP, PR, RR.

Examination of R leg:

Patient in supine position.

On Inspection,

Attitude: Hip - externally rotated.

Knee - mid flexion

Ankle - mid plantar flexion & inverted

- Deformity present at the junction of mid & lower 1/3rd of leg, which is bent. anterolaterally (deviation to which side
↓
varus/valgus/tort)

- Irregular in shape.

- Skin over deformity (N).

- No scars, sinuses, dilated veins, ulcers.

- No wasting of muscles.

- Apparent shortening of limb.

- Gait: Able to stand $\bar{=}$ support.

Unable to walk.

On palpation,

- not warm, no tenderness.

- A single (swelling) (angulation) present at junction of middle & lower 1/3rd of leg.

apex - anterolaterally (anteromedially)

Irregular in shape

Size of 7x8 cms

- Broadening & thickening of bone present.

- Skin over deformity (N).
- Hard in consistency, smooth surface.
- No abnormal mobility
- No crepitus. $\rightarrow$ No $\oplus$ Transmitted mobility.

Measurements:

Thigh

Leg

 $\mathbb{R}$

48 cm

34 ans

④ .

48 cms

38 cms .

$\text{Limb length} = \text{Leg length} + \text{discrepancy of } 4\text{ms}$ corresponding to leg segment.

Movements:

- Movements:
- At ankle joint: Active - All restricted
↓ plantar flexion
↓ Dorsiflexion, ↓ Inversion & eversion.

- Distal neurovascular status: (N).
- No LN enlargement.
- other joints (N) . , opposite limb (N).

Diagnosis:

other joints (N) . , opposite

Diagnosis:

A case of malunited / maluniting # of both bones of ^{say no bone} (R) leg
at junction of middle & lower 1/3rd of leg, with shortening
of (R) limb & anterolateral angulation.

Inv: x-ray.

Inv: X-ray.

Rx: Mild - no Rx needed (no disability, expected to correct by remodelling).

If disability present,

☆ Osteoclasia (res bone) → mild to moderate angular deformities in children. (→ immobilisation).
→ internal fixation

(★) Redoing # surgically (M/c) → open → c/t melunion → internal fixation
bone grafting.
Isolated bone → osteotomy at site

(★) Correcting osteotomy → if poor skin & vascularity of bone ⇒ osteotomy at site away from #.

(*) Excision of protruding bone

[Corrective osteotomy (1) + Bone graft (3) + Internal fixation (2)]